

CIS 4270/5270: Trustworthy Machine Learning Spring 2026: Course Project Proposal

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Contributions:

Reinforcement Learning Task

We propose a constraint-following recipe generation task. Given a list of available ingredients, the model must generate a quick (5-10 minute), student-friendly recipe using those ingredients.

Training Data

We will generate synthetic data using a stronger model (GPT-5 mini). Each sample will include:

- an ingredient list (e.g., eggs, bread, cheese, onion),
- a prompt specifying “quick student-friendly recipe,”
- and a recipe satisfying these constraints.

We will also generate negative examples that violate constraints to support preference learning.

Reward Function

Our reward function combines automatic checks and learned evaluation:

- **Ingredient usage:** Reward if the recipe uses provided ingredients; penalize unrelated ingredients.
- **Simplicity:** Reward short, concise recipes with fewer steps; penalize long procedures. Small bonus for N steps in [min, max] to avoid one-line recipes.
- **Structure:** Reward clear, step-by-step formatting
- **Preference satisfaction:** Use GPT-5 mini (LLM-as-judge) to score whether the recipe is practical and student-friendly

Training will use a continuous reward with partial credit (for partially satisfying constraints).

Evaluation will report both:

- binary constraint satisfaction rate (whether all constraints are fully met)
- ingredient compliance rate
- average recipe length (based on number of steps and average cooking time)
- preference satisfaction score (via an LLM-as-judge)

DPO Data (Preference Pairs)

We plan to construct preference pairs using a stronger model such as GPT-5 mini:

- **Preferred:** Recipes that satisfy constraints (correct ingredients, concise, well-structured)
- **Rejected:** Recipes that violate constraints (unavailable ingredients, too complex/time-consuming, unclear steps)

Collaboration plan:

- **Akshat:** RFT, reward design, evaluation
- **Rashi:** SFT, data generation
- **Ritika:** DPO, experiments, analysis

All members will contribute to the final report and presentation.